

Linear Algebra – MAT 2610

Section 4.6 (Change of Basis)

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Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for V and x is in V . We have seen how to find $[x]_{\mathcal{B}}$ the coordinates of x relative to $\mathcal{B}$. We have also seen $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$ using the change of coordinates matrix.

We also know that a basis is not unique. So if $\mathcal{C} = \{c_1, \dots, c_n\}$ is another basis of V , we can also find $[x]_{\mathcal{C}}$ and $P_{\mathcal{C}}$.

But what if we have $[x]_{\mathcal{B}}$ and we want to find $[x]_{\mathcal{C}}$? We need a matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ to go directly from $[x]_{\mathcal{B}}$ to $[x]_{\mathcal{C}}$.



We have

$$\underline{x = P_B [x]_B}$$

Also

$$x = P_C [x]_C \quad \leftarrow$$

$$P_C^{-1} x = [x]_C \quad \leftarrow$$

$$P_C^{-1} (P_B [x]_B) = [x]_C$$

$$\underline{(P_C^{-1} P_B)} \underline{[x]_B} = [x]_C$$

So the matrix $\underline{P_{C \leftarrow B}} = \underline{P_C^{-1} P_B}$ transforms $[x]_B$ into $[x]_C$



Now, consider $P_{C \leftarrow B} = P_C^{-1} P_B$. Recall that P_C^{-1} is the matrix that transforms a vector in the standard coordinates into a vector in the C coordinates.

$$P_C^{-1} x = [x]_C$$

But what is P_B ? This is the matrix made up of the vectors of the basis $B = \{b_1, \dots, b_n\}$. Each of these basis elements is itself a vector in the standard coordinates.

$$P_B = \begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix}$$

$\uparrow$

Hence, every column of $P_{C \leftarrow B}$ is the translation of a basis vector of the basis $B = \{b_1, \dots, b_n\}$ into C coordinates.

$$P_{C \leftarrow B} = \begin{bmatrix} P_C^{-1} b_1 & P_C^{-1} b_2 & \dots & P_C^{-1} b_n \end{bmatrix}$$



Theorem 15

Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases for a vector space V . Then there is a unique $n \times n$ matrix $P_{\mathcal{C} \leftarrow \mathcal{B}}$ such that

$$[x]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}} [x]_{\mathcal{B}}$$

The columns of $P_{\mathcal{C} \leftarrow \mathcal{B}}$ are the $\mathcal{C}$ -coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = [\underbrace{[b_1]_{\mathcal{C}}} \dots [b_n]_{\mathcal{C}}]$$



Example

Let $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$ and $\mathcal{C} = \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$.

If a vector v has coordinates $[v]_{\mathcal{C}} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, find $[v]_{\mathcal{B}}$



Solution

We know:

$$\underline{x} = P_B[v]_B \text{ where } P_B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\underline{x} = P_C[v]_C \text{ where } P_C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\text{We have } [v]_C = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \text{ so } P_B[v]_B = P_C[v]_C$$

$$\text{Or } \underline{[v]_B} = \underline{P_B^{-1} P_C [v]_C} = \underline{P_{B \leftarrow C} [v]_C}$$



Solution

$$P_B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \text{ so } P_B^{-1} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$P_{B \leftarrow C} = P_B^{-1} P_C = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{4}{5} \\ 0 & \frac{3}{5} \end{bmatrix}$$

$$\text{So } [v]_B = \begin{bmatrix} 1 & \frac{4}{5} \\ 0 & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{27}{5} \\ \frac{9}{5} \end{bmatrix} \quad \star$$

$$v = P_C [v]_C$$

$$v = P_B [v]_B$$



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